CDAC

NVIDIA vGPU Installation, Configuration & DLS Licensing Guide

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Document: Installation, Configuration and DLS Licensing | Platform: VMware vSphere 8.0

1. Purpose and Scope

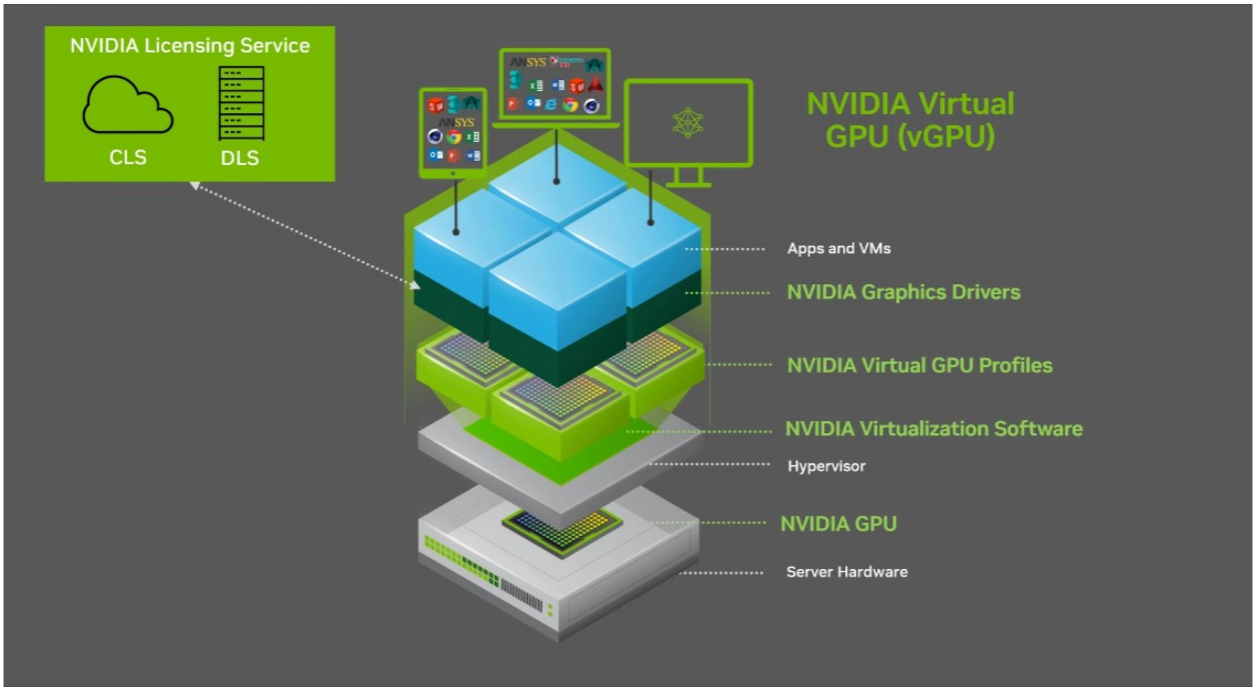
This document provides a step-by-step procedure for installing and configuring NVIDIA vGPU on VMware vSphere 8.0, configuring a virtual machine with a vGPU profile, installing the NVIDIA guest driver and CUDA toolkit, and configuring NVIDIA licensing through CLS/DLS. The technical commands, values, and workflow are retained from the supplied source document.

2. Source Prerequisites

NVIDIA portal access is required when licensing has been purchased.
• Product Family: vGPU or NVAIE
• Platform: VMware vSphere
• Platform Version: 8.0
• Download the latest applicable version from the available list.
After download, the package contains the VM/Guest Driver and Host Driver components.

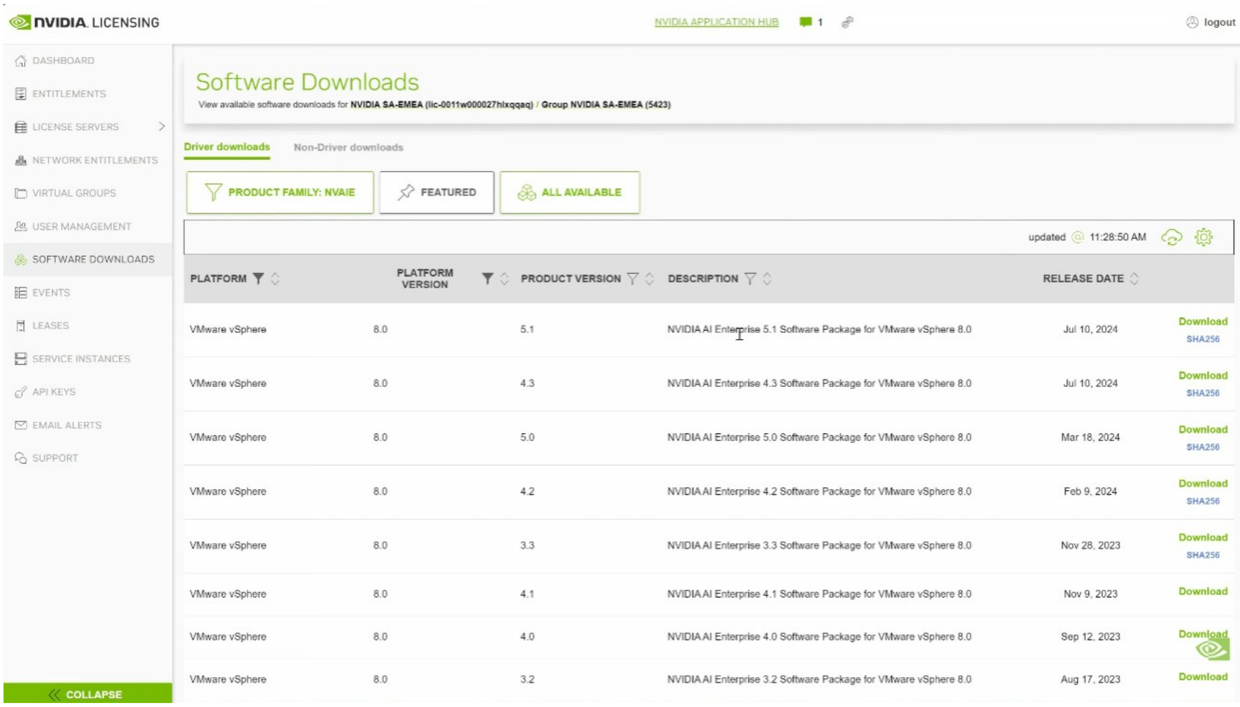
3. Document Contents

1. NVIDIA vGPU Host Driver Installation
2. vCenter Configuration
3. VM Creation and vGPU Assignment
4. Linux Guest OS vGPU Driver Installation
5. Understanding vGPU Profiles
6. NVIDIA CLS / DLS Licensing
7. DLS High Availability (HA)
8. Client Configuration Token and Validation



Download nvidia driver bundle package

If you have purchased the licensing, you will be able to access this page using your ID and password.



Select –
PRODUCT FAMILY: vGPU or NVAIE
Platform: VMware vSphere
Platform Version: 8.0
Now download the latest version from the list.

After downloading, we will get the files below, which contain the VM driver and Host Driver.

Guest_Drivers	7/30/2024 11:42 AM	File folder	
Host_Drivers	7/30/2024 11:42 AM	File folder	
Signing_Keys	7/30/2024 11:42 AM	File folder	
550.90.05-550.90.07-552.74-nvidia-ai-enterprise-product-support-matrix.pdf	7/30/2024 11:41 AM	Microsoft Edge PDF ...	271 KB
550.90.05-550.90.07-552.74-nvidia-ai-enterprise-quick-start-guide.pdf	7/30/2024 11:41 AM	Microsoft Edge PDF ...	748 KB
550.90.05-550.90.07-552.74-nvidia-ai-enterprise-release-notes.pdf	7/30/2024 11:41 AM	Microsoft Edge PDF ...	478 KB
550.90.05-550.90.07-552.74-nvidia-ai-enterprise-user-guide.pdf	7/30/2024 11:41 AM	Microsoft Edge PDF ...	3,440 KB

Guest Driver and License Server Files

We will get the NVIDIA driver and License Server files under the Guest Driver folder.

Guest_Drivers   Search Guest_Drivers

Name	Date modified	Type	Size
552.74_grid_win10_win11_server2022_dch_6...	7/30/2024 11:42 AM	Application	492,993 KB
nvidia-linux-grid-550_550.90.07_amd64.deb	7/30/2024 11:42 AM	DEB File	344,420 KB
nvidia-linux-grid-550-550.90.07-1.x86_64.rpm	7/30/2024 11:42 AM	RPM File	453,780 KB
NVIDIA-Linux-x86_64-550.90.07-grid.run	7/30/2024 11:42 AM	RUN File	305,311 KB

Host Driver File

Under the Host Driver folder, we will find the ESXi driver file.

Host Drivers ›   Search Host_Drivers

Name	Date modified	Type	Size
NVD-AIE_ESXi_8.0.0_Driver_550.90.05-1OEM.800.1.0.20613240.vib	7/30/2024 11:42 AM	VIB File	109,135 KB
NVD-AIE_800_550.90.05-1OEM.800.1.0.20613240_23968270.zip	7/30/2024 11:42 AM	Compressed (zipped)...	108,155 KB
nvd-gpu-mgmt-daemon_550.90.05-0.0.0000_23967827.zip	7/30/2024 11:42 AM	Compressed (zipped)...	382 KB

Now copy all the required files to the server.

Host Prerequisite Checks

Check these settings on the server; if they are not enabled, enable them first.

CHECK YOUR BIOS SETTINGS FIRST

Intel VT-D **AMD IOMMU**	**Memory Mapped I/O above 4 GB** **Above 4G Decoding** **Above 4GB MMIO**	**Ampere GPUs and above:** **SR-IOV Support** **SR-IOV Global Enable**

Enter Maintenance Mode and Check Existing Installation

esxcli system maintenanceMode set --enable true

nvidia-smi

```
[root@dc2xtmvmesx01-vsan1:~] esxcli system maintenanceMode set --enable true
[root@dc2xtmvmesx01-vsan1:~] nvidia-smi
-sh: nvidia-smi: not found
[root@dc2xtmvmesx01-vsan1:~] esxcli software vib list |grep NV
[root@dc2xtmvmesx01-vsan1:~]
```

→ If installed then we get below error

```
[root@dc2xtmvmesx01-vsan1:~] esxcli software vib update -d /vmfs/volumes/SSD2/vib/
NVD-AIE-800 535.183.04-1OEM.800.1.0.20613240 23954161.zip  nvd-gpu-mgmt-daemon 535.183.04-0.0.0000 23959104.zip
NVD-AIE-800_550.90.05-1OEM.800.1.0.20613240_23968270.zip   nvd-gpu-mgmt-daemon_550.90.05-0.0.0000_23967827.zip
[root@dc2xtmvmesx01-vsan1:~] esxcli software vib update -d /vmfs/volumes/SSD2/vib/NVD-AIE-800_550.90.05-1OEM.800.1.0.20613240_23968270.zip
 [LiveInstallationError]
 NVD_bootbank_NVD-AIE_ESXi_8.0.0_Driver_535.183.04-1OEM.800.1.0.20613240: Failed to unmount tardisk nvd_boot.v00 of VIB NVD_bootbank_NVD-AIE_ESXi_8.0.0_Driver_535.183.04
00.1.0.20613240: Error in running [rm /tardisks/nvd_boot.v00]:
 Return code: 1
 Output: rm: can't remove '/tardisks/nvd_boot.v00': Device or resource busy

 It is not safe to continue. Please reboot the host immediately to discard the unfinished update.
      cause = ('NVD-AIE-800(535.183.04-1OEM.800.1.0.20613240)', "Failed to unmount tardisk nvd_boot.v00 of VIB NVD_bootbank_NVD-AIE_ESXi_8.0.0_Driver_535.183.04-1OEM.800
613240: Error in running [rm /tardisks/nvd_boot.v00]:\nReturn code: 1\nOutput: rm: can't remove '/tardisks/nvd_boot.v00': Device or resource busy\n")
       vibs = ['NVD_bootbank_NVD-AIE_ESXi_8.0.0_Driver_535.183.04-1OEM.800.1.0.20613240']
 Please refer to the log file for more details.
[root@dc2xtmvmesx01-vsan1:~]
```

To remove this condition, uninstall the listed NVIDIA package and stop the associated daemon service. First, check the list of installed software using the command below.

Command : esxcli software vib list | grep -i nvidia

```
[root@dc2xtmvmesx01-vsan1:~] esxcli software vib list |grep NVD
NVD-AIE_ESXi_8.0.0_Driver      535.183.04-1OEM.800.1.0.20613240      NVD     VMwareAccepted     2024-07-29
nvdgpumgmtdaemon               535.183.04-1OEM.700.1.0.15843807      NVD     VMwareAccepted     2024-07-29
[root@dc2xtmvmesx01-vsan1:~]
```

Remove the software using the command below.

Command: esxcli software vib remove listed_software

Install NVIDIA Host Software

esxcli software vib install -d /path-of-zip-file

For a VIB package, use the VIB installation command supplied for that release. Example:

esxcli software vib install -v /vmfs/volumes/<datastore>/NVIDIA/<NVIDIA-vGPU-VMware_ESXi-*.vib>

```
[root@dc2xtmvmesx01-vsan1:~] esxcli software vib install -d /vmfs/volumes/SSD2/vib/NVD-AIE-800_550.90.05-1OEM.800.1.0.20613240_23968270.zip
Installation Result
   Message: Operation finished successfully.
   Reboot Required: false
   VIBs Installed: NVD_bootbank_NVD-AIE_ESXi_8.0.0_Driver_550.90.05-1OEM.800.1.0.20613240
   VIBs Removed:
   VIBs Skipped:
[root@dc2xtmvmesx01-vsan1:~] esxcli software vib install -d /vmfs/volumes/SSD2/vib/nvd-gpu-mgmt-daemon_550.90.05-0.0.0000_23967827.zip
Installation Result
   Message: The update completed successfully, but the system needs to be rebooted for the changes to be effective.
   Reboot Required: true
   VIBs Installed: NVD_bootbank_nvdgpumgmtdaemon_550.90.05-1OEM.700.1.0.15843807
   VIBs Removed:
   VIBs Skipped:
[root@dc2xtmvmesx01-vsan1:~]
```

Reboot

reboot

After reboot, verify the NVIDIA module and GPU again:
Verify the vGPU daemon.

Command : /etc/init.d/nvdGpuMgmtDaemon status

```
[root@dc2xtmvmesx01-vsan1:~] /etc/init.d/nvdGpuMgmtDaemon status
daemon_nvdGpuMgmtDaemon is running
[root@dc2xtmvmesx01-vsan1:~]
```

vmkload_mod -l | grep nvidia
esxcli software vib list | grep -i nvidia
Nvidia-smi

```
[root@dc2xtmvmesx01-vsan1:~] nvidia-smi
Tue Jul 30 10:59:52 2024
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 550.90.05              Driver Version: 550.90.05      CUDA Version: N/A      |
|-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA L4                      On  |   00000000:02:00.0 Off |                    0 |
| N/A   37C    P8             17W /   72W |       0MiB /  23034MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+
|   1  NVIDIA L40                     On  |   00000000:84:00.0 Off |                    0 |
| N/A   28C    P8             41W /  300W |       0MiB /  46068MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI        PID   Type   Process name                              GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
[root@dc2xtmvmesx01-vsan1:~]
```

Exit maintenance mode.

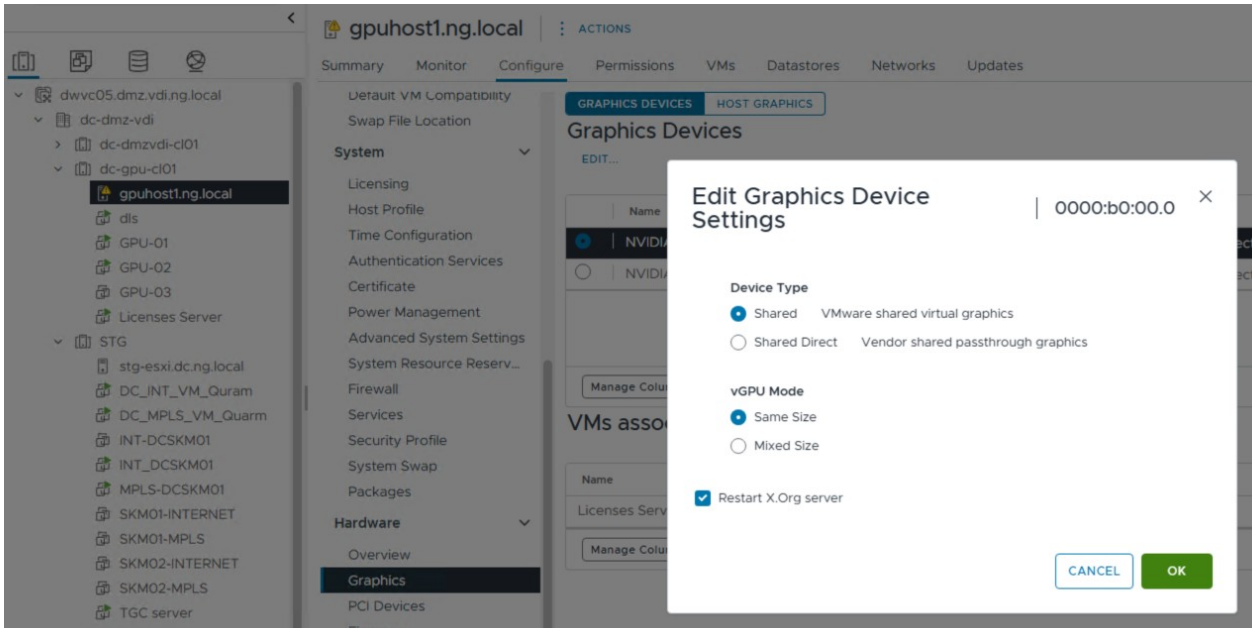
esxcli system maintenanceMode set --enable false

Installation Done

4. vCenter Configuration

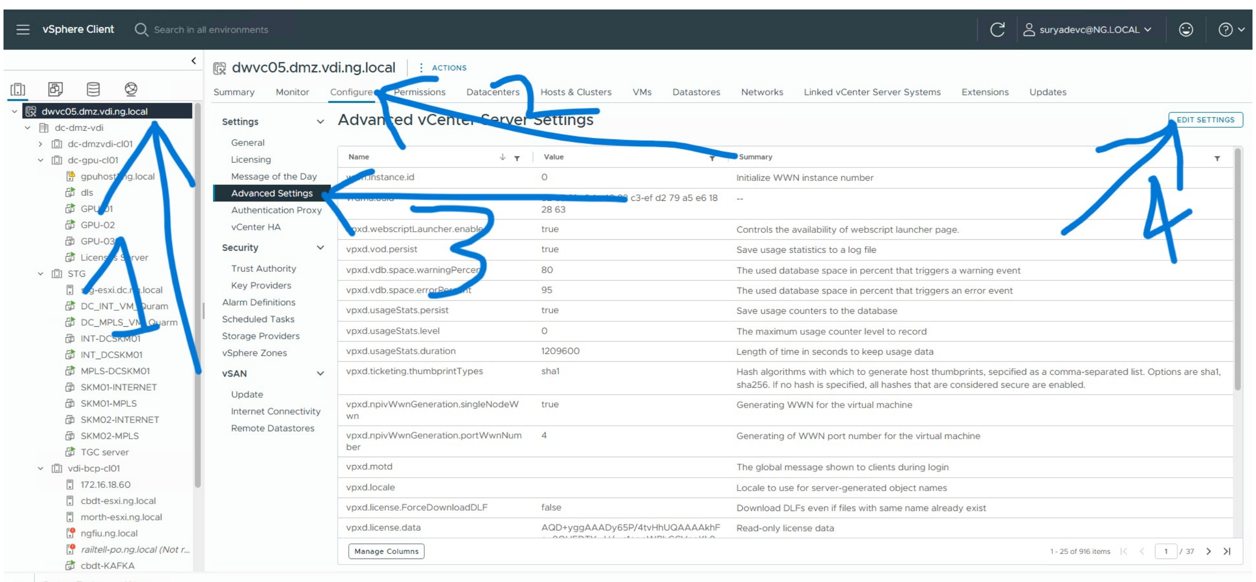
Steps:-

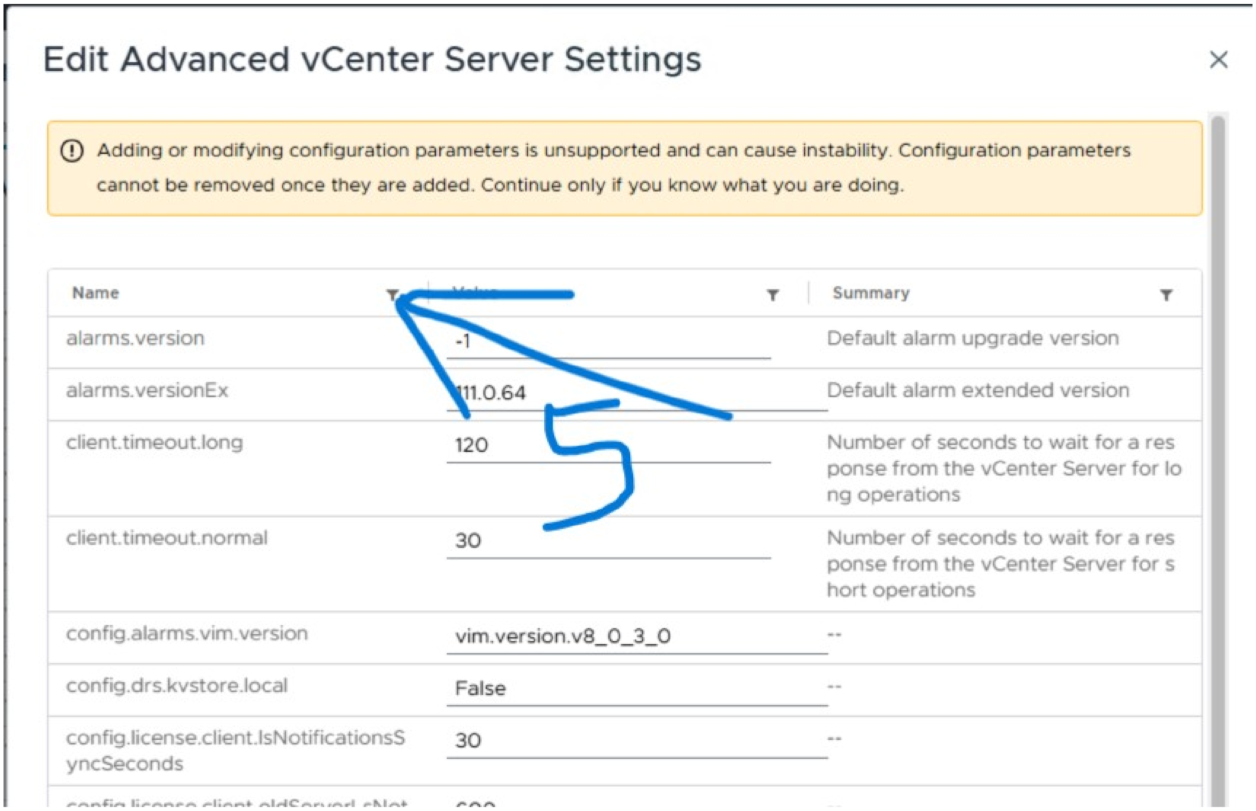
1. Open vCenter > Select your server > Configure > Graphics > HOST GRAPHICS > Edit and make sure you select ‘Shared’ as shown below.

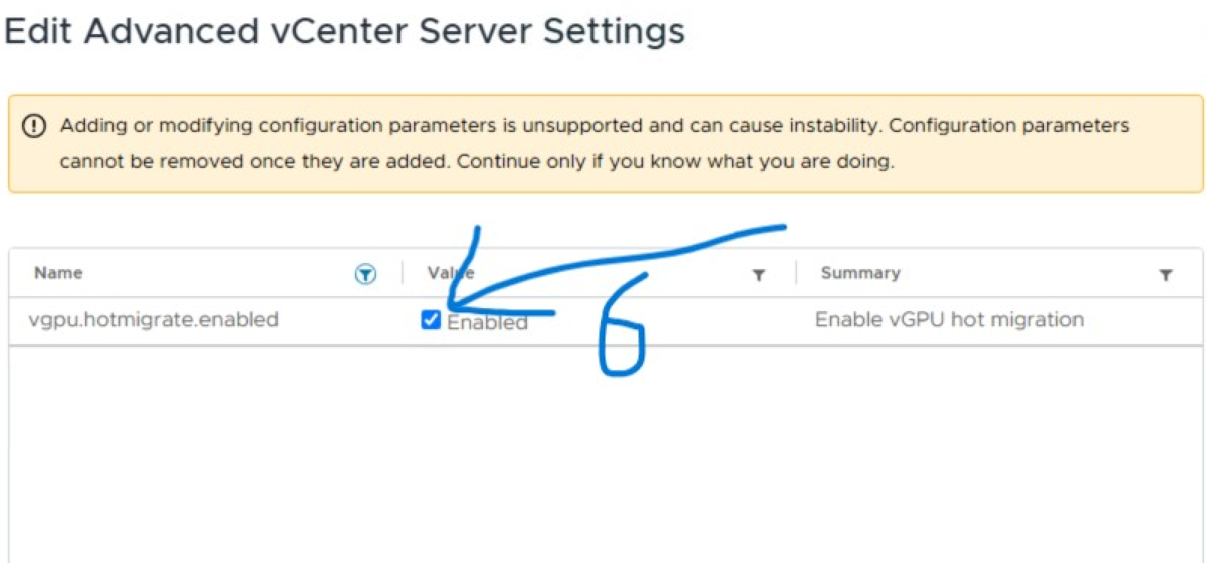


Enable the vGPU Advanced Setting

2. Open Vcenter > Select Your data center from top of inventory hierarchy > Configure > Advanced setting > Edit setting > Now filter on name section and type vgpu > Enable setting by check box like below screenshot







Save the setting.

Vcenter Configuration complete

5. VM Creation and vGPU Configuration

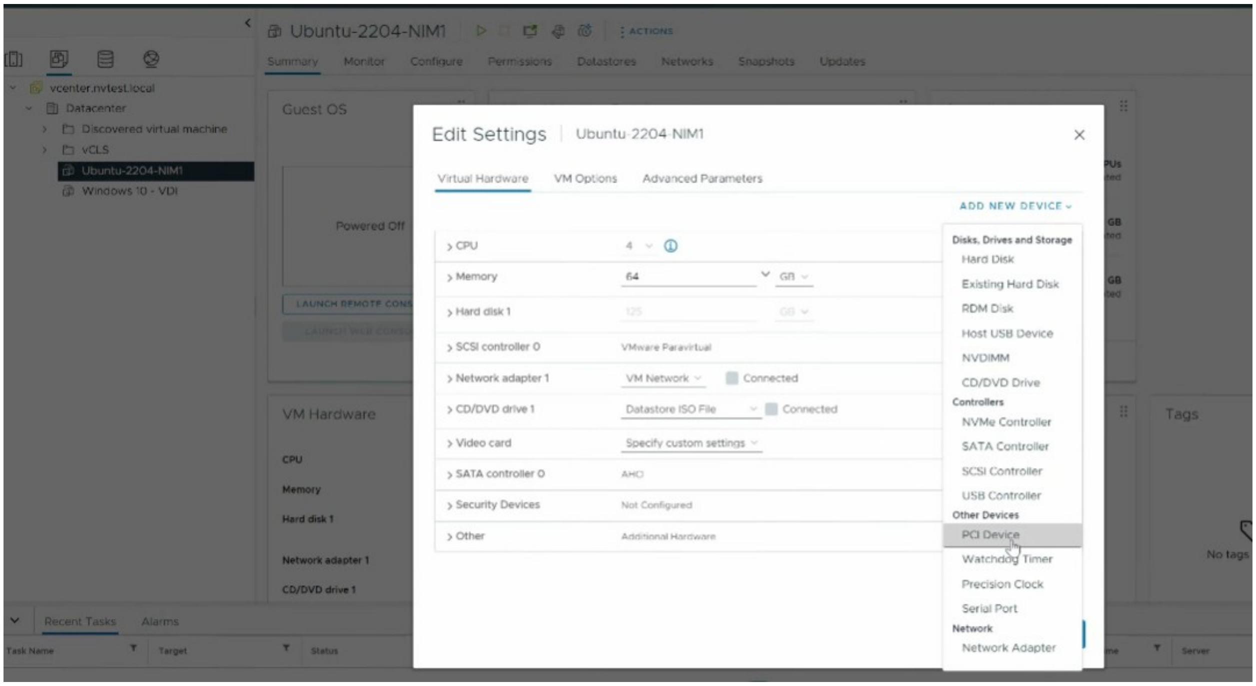
Steps:

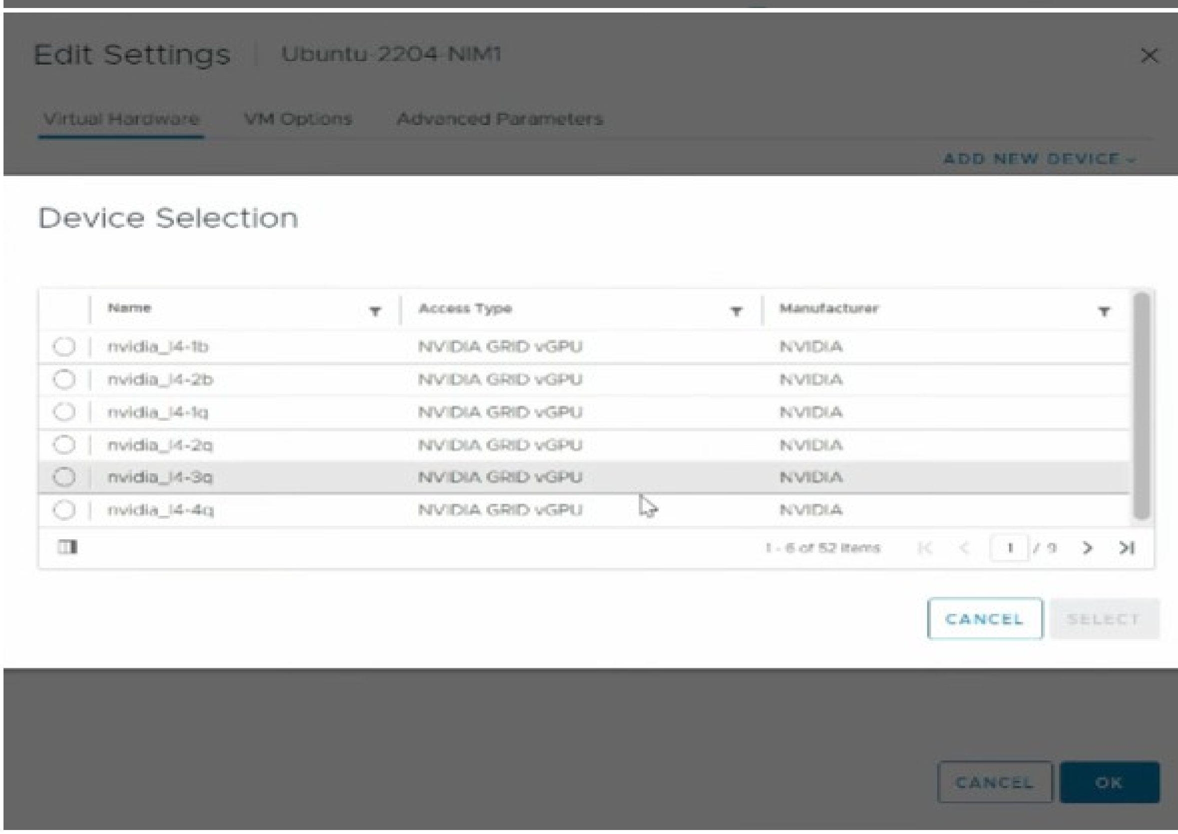
Create a VM Using a Supported OS

1. Create a VM using a supported OS and according to your required configuration.

Add a vGPU Profile to the VM

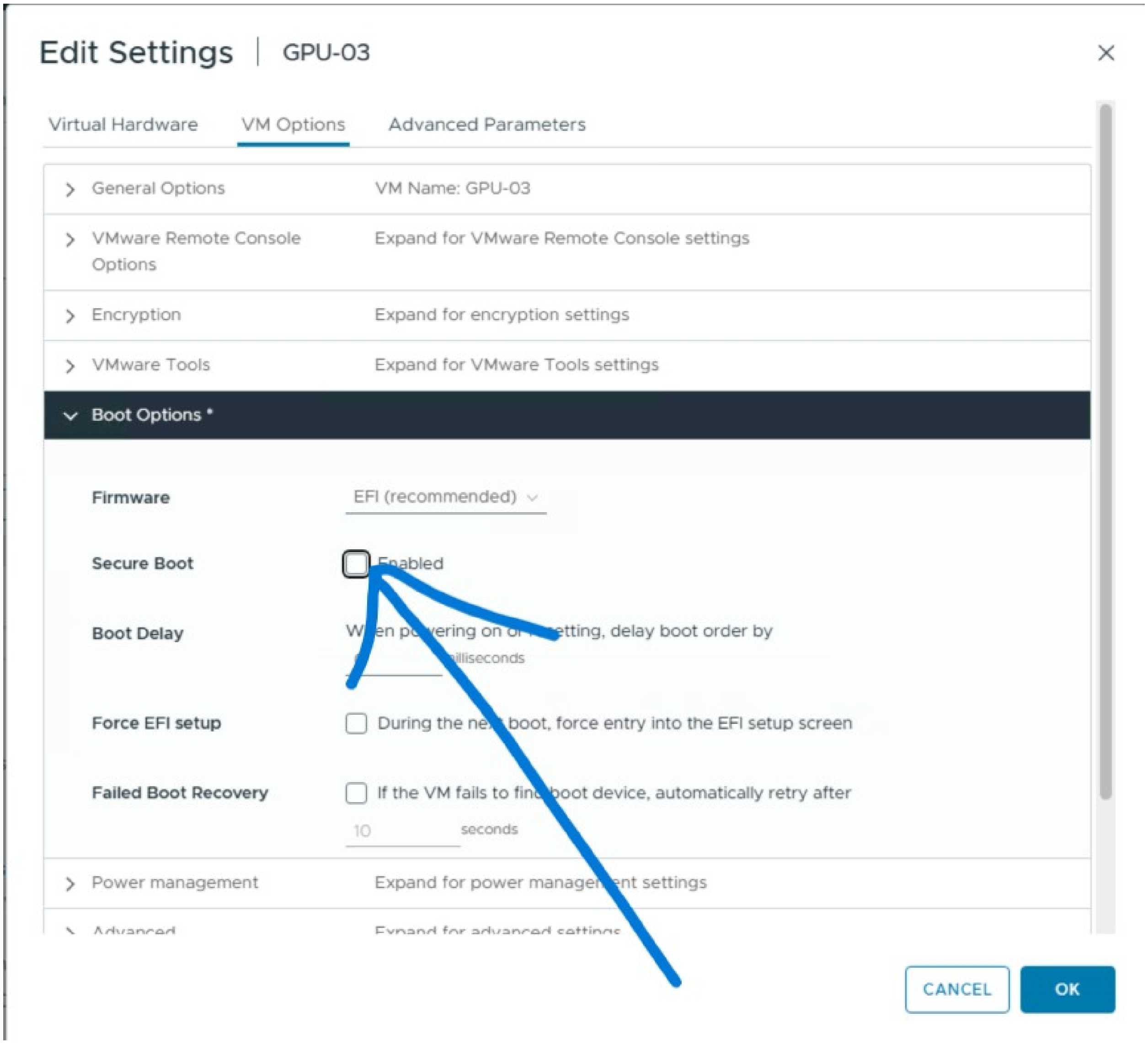
2. To add vGPU to the VM, select VM > Edit Settings > Add PCI Device > Select the vGPU profile according to your requirement.



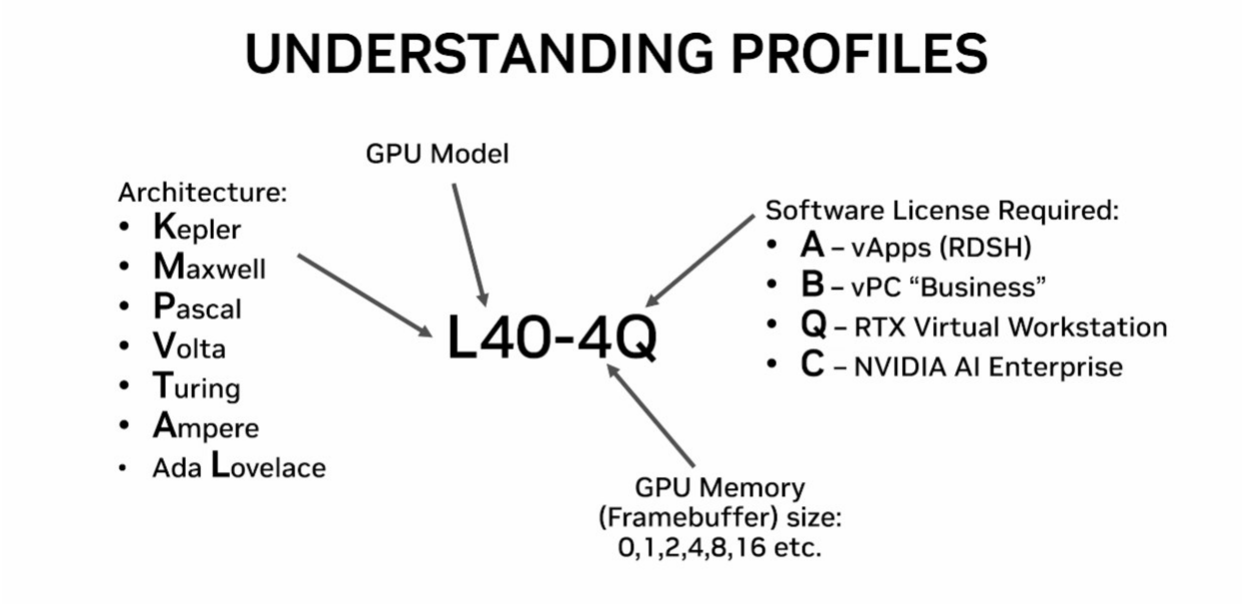


Disable Secure Boot

3. Disable Secure Boot.
   Select VM > Edit Settings > VM Options > Boot Options > Uncheck Secure Boot


Edit Settings
GPU-03
Virtual Hardware
VM Options
Advanced Parameters
General Options
VM Name: GPU-03
VMware Remote Console Options
Expand for VMware Remote Console settings
Encryption
Expand for encryption settings
VMware Tools
Expand for VMware Tools settings
Boot Options *
Firmware
EFI (recommended)
Secure Boot
Boot Delay
Force EFI setup
Failed Boot Recovery
If the VM fails to find boot device, automatically retry after
10
seconds
Power management
CANCEL
OK

6. Understanding vGPU Profiles



7. Linux Guest OS vGPU Installation

Steps:-

Disable Nouveau

1. `blacklist nouveau`   # It responsible to deny load of linux Nouveau NVIDIA driver
2. `options nouveau modeset=0`   # It responsible to disable kernel modesetting ouveau driver

   :wq!

3. Regenerate initramfs

→ dracut --force

4. Stop and Disable the GDM Service

5. `systemctl disable gdm.service`
6. systemctl disable gdm.service

7. Configure the Local Package Source

8. Reboot the Guest OS

→ reboot

9. Install Required Build and Kernel Packages

→ yum install gcc make kernel-devel-$(uname -r) kernel-headers-$(uname -r) dkms kernel-devel-matched-$(uname -r)

10. Copy and Install the NVIDIA vGPU Guest Driver

→ rpm -ivh nvidia-linux-grid-550-550.90.07-1.x86_64.rpm

11. Verify the NVIDIA Driver Installation

→ nvidia-smi

If the installation is successful, you should see the output shown below.

```
root@ubuntu-2204-nim1:/home/lee# nvidia-smi
Tue Jul 30 14:41:50 2024
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 550.90.07              Driver Version: 550.90.07      CUDA Version: 12.4     |
|-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA L40-48C                 On  |   00000000:02:03.0 Off |                    0 |
| N/A   N/A    P8              N/A /  N/A |      52MiB /  49152MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI        PID   Type   Process name                              GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|    0   N/A  N/A      1027      G   /usr/lib/xorg/Xorg                             51MiB |
+-----------------------------------------------------------------------------------------+
```

12. Download and Install the Supported CUDA RPM

→ rpm -ivh cuda-repo-rhel9-13-2-local-13.2.0_595.45.04-1.x86_64.rpm

```
[root@GPU-01 ClientConfigToken]# cd /etc/yum.repos.d/
[root@GPU-01 yum.repos.d]# ll
total 8
-rw-r--r--. 1 root root 190 Mar  6 20:50 cuda-rhel9-13-2-local.repo
-rw-r--r--. 1 root root 288 Aug 17 19:07 local.repo
-rw-r--r--  1 root root   0 Aug 17 19:45 redhat.repo
[root@GPU-01 yum.repos.d]# cat cuda-rhel9-13-2-local.repo
[cuda-rhel9-13-2-local]
name=cuda-rhel9-13-2-local
baseurl=file:///var/cuda-repo-rhel9-13-2-local
enabled=1
gpgcheck=1
gpgkey=file:///var/cuda-repo-rhel9-13-2-local/913BAD1A.pub
obsoletes=0
```

```
dnf clean all
```

dnf install cuda-toolkit-13-2

13. Configure /etc/nvidia/gridd.conf

`ServerAddress=X.X.X.X# License server IP`

FeatureType=4#For virtual workstation set 2 but for nvidia vgpu must should be 4

14. Copy the Client Configuration Token and Restart the Service

→ service nvidia-gridd restart

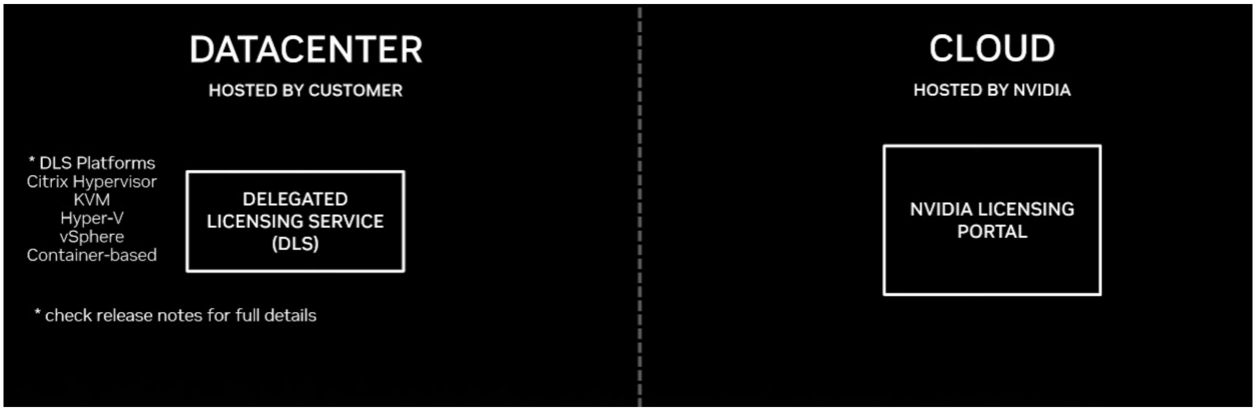
```
[root@localhost ~]# cp client_configuration_token_08-17-2026-15-03-00.tok /etc/nvidia
/ClientConfigToken/
[root@localhost ~]# cd /etc/nvidia/ClientConfigToken/
[root@localhost ClientConfigToken]# ls
client_configuration_token_08-17-2026-15-03-00.tok
[root@localhost ClientConfigToken]#
[root@localhost ClientConfigToken]# service nvidia-gridd restart
Redirecting to /bin/systemctl restart nvidia-gridd.service
[root@localhost ClientConfigToken]# nvidia-smi -q | more
```

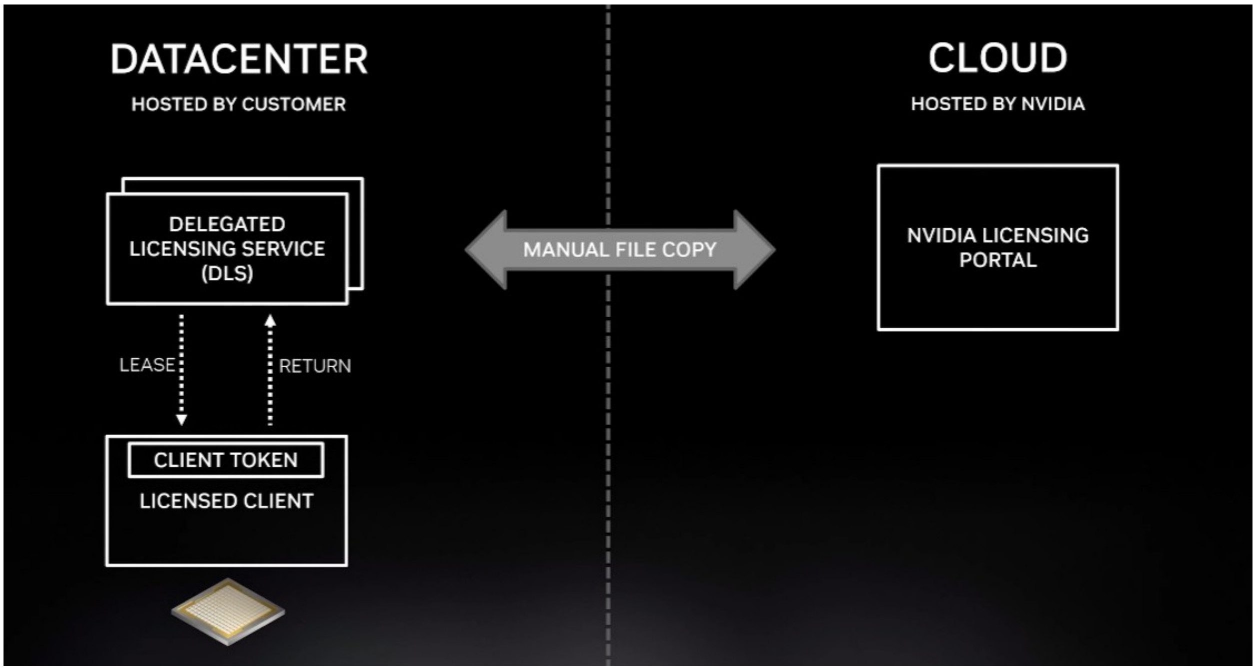
```
        vGPU Heterogeneous Mode                           : N/A
    vGPU Software Licensed Product
        Product Name                                      : NVIDIA vGPU for Compute
        License Status                                    : Licensed (Expiry: 2026-8-18
9:35:55 GMT)
    GPU Recovery Action                                   : None
    GSP Firmware Version                                  : N/A
    IBMNPU
        Relaxed Ordering Mode                             : N/A
    PCI
        Bus                                               : 0x03
```

GPU server setup completed

8. NVIDIA Licensing: CLS and DLS

CLS – Cloud License Server
DLS – Delegated License Server
CLS: The Cloud License Server is hosted by NVIDIA and can be accessed online.
DLS: The Delegated License Server is the closest equivalent to the previous on-premises licensing service. It is a virtual appliance that can be downloaded and installed by the customer. It supports VMware, Citrix, Hyper-V, and KVM, as well as container-based deployment with Docker. NVIDIA also provides deployment options through Kubernetes, OpenShift, and Tanzu.
DLS can also be configured for HA, and the version can be updated quickly.
DLS communicates with the licensing service through manual file transfer; there is no automatic synchronization between the NVIDIA licensing service and the NVIDIA licensing portal.


DATACENTER
HOSTED BY CUSTOMER
* DLS Platforms
Citrix Hypervisor
KVM
Hyper-V
vSphere
Container-based
DELEGATED LICENSING SERVICE (DLS)
* check release notes for full details
CLOUD
HOSTED BY NVIDIA
NVIDIA LICENSING PORTAL


DATACENTER
HOSTED BY CUSTOMER
DELEGATED LICENSING SERVICE (DLS)
LEASE
RETURN
CLIENT TOKEN
LICENSED CLIENT
MANUAL FILE COPY
CLOUD
HOSTED BY NVIDIA
NVIDIA LICENSING PORTAL

8.1 Create and Configure CLS

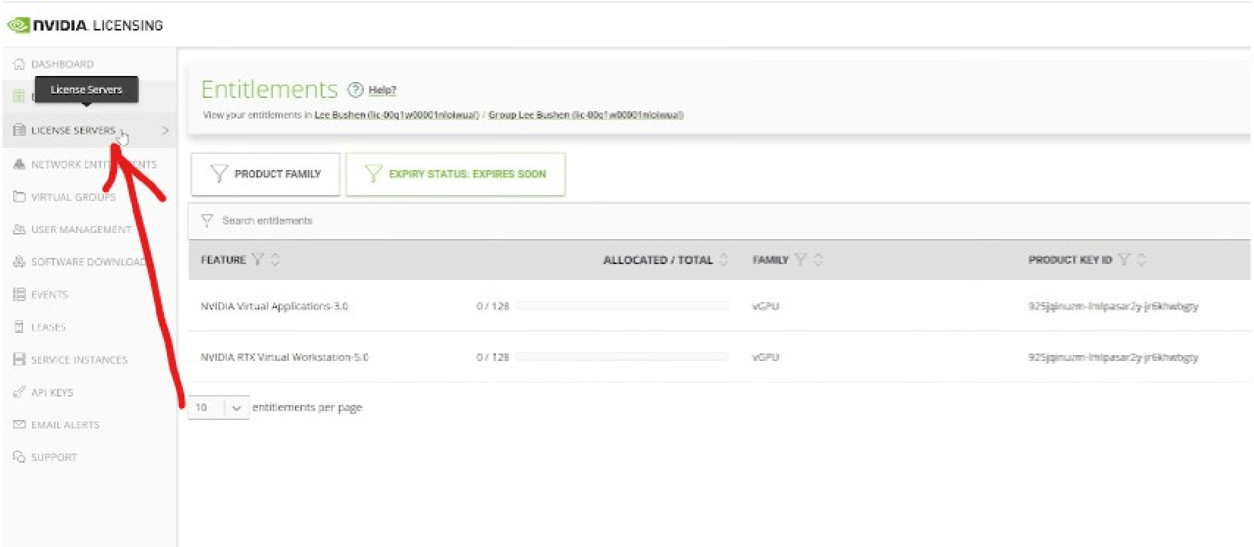
Steps

1. **Log in to the NVIDIA Portal**

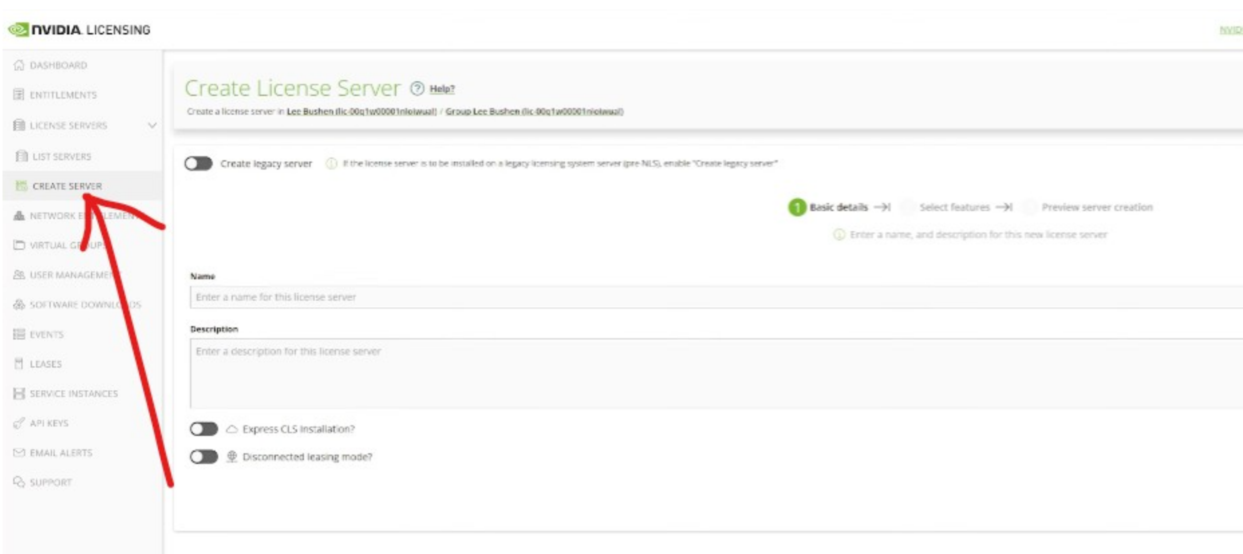
2. **Open License Server**

3. **Create Server**

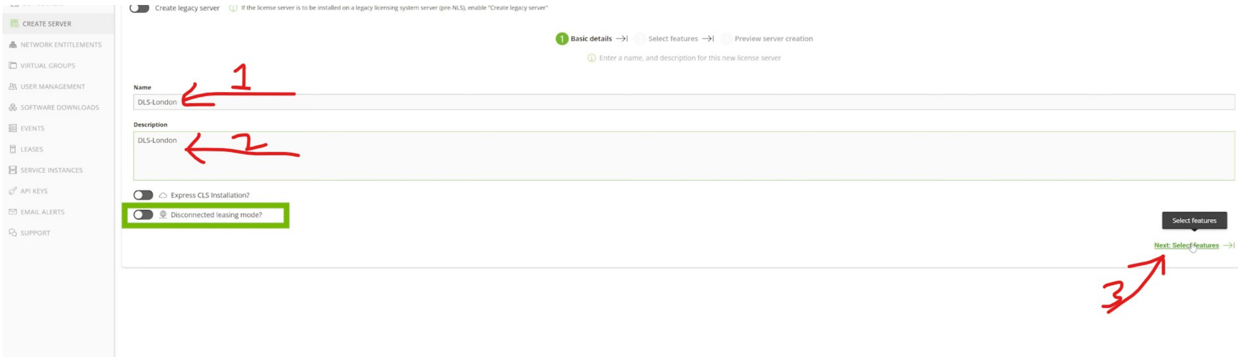
   Enter Server Name and Description



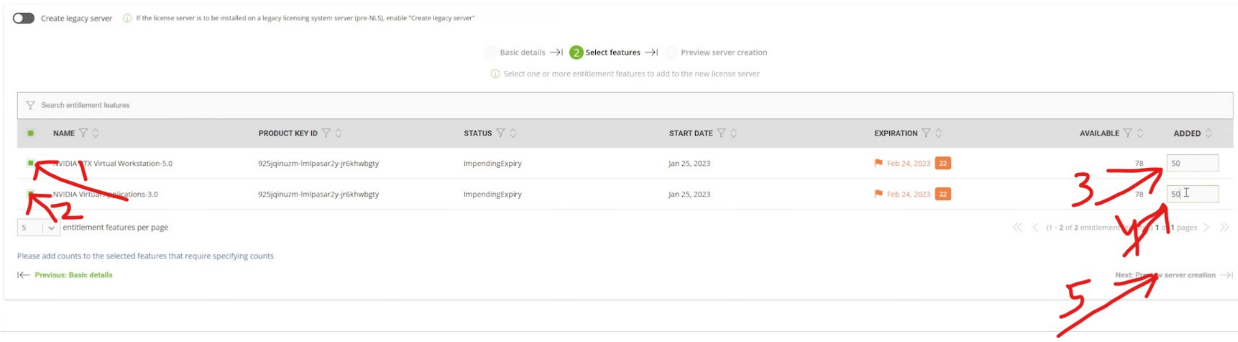
4. Create server



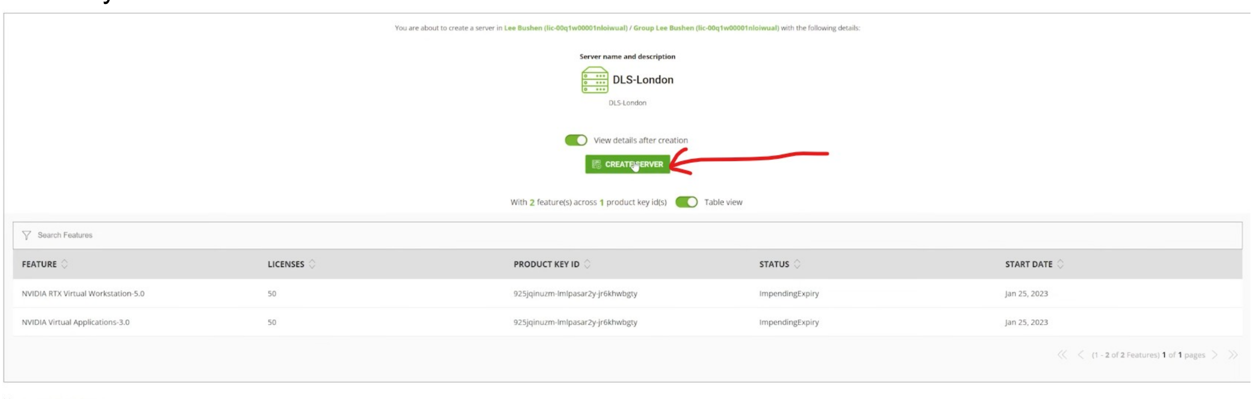
5. Enter the server name and description, then click Next.



6. Select the Required Feature and click Next.



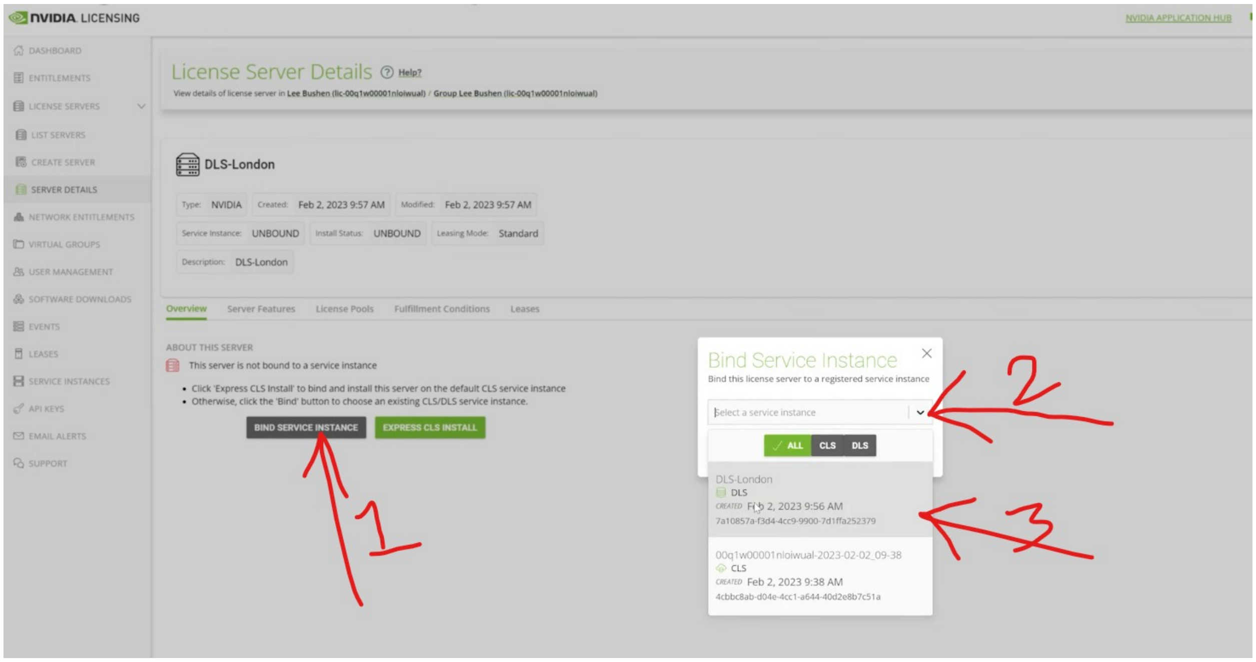
7. Review your server details and click on ‘CREATE SERVER’.



CLS server created…

8. Bind the CLS Service Instance to license server

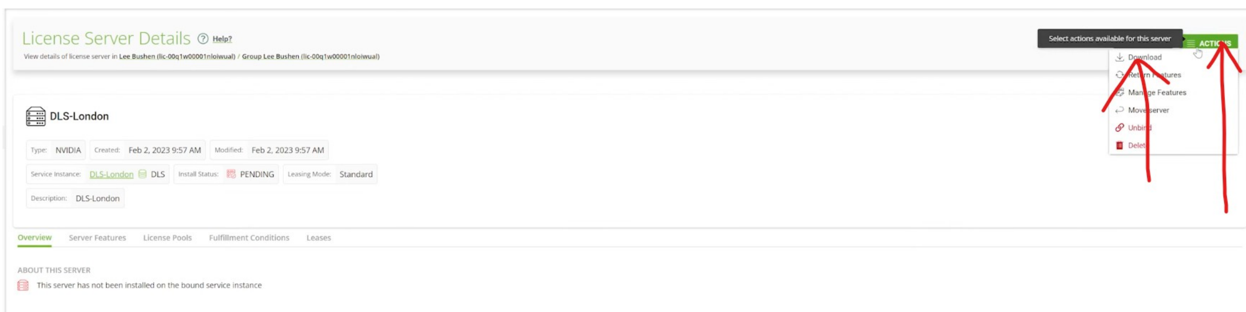
Bind service instance > select created CLS server > Bind



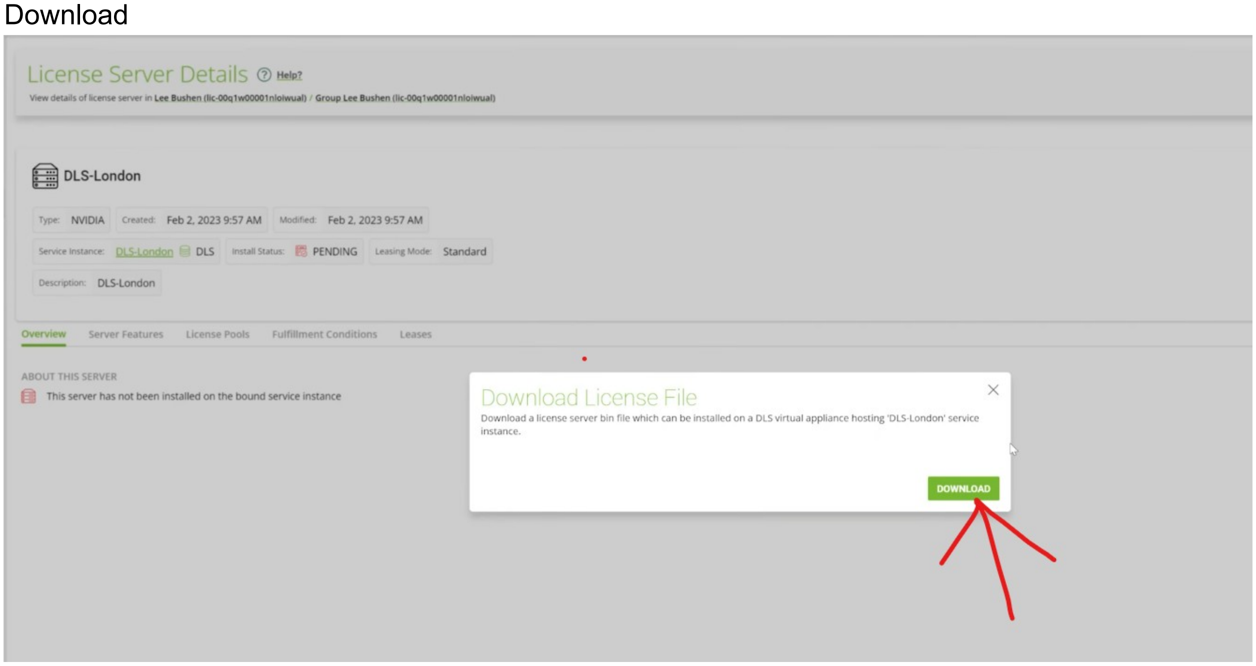
License server bind successfully

9. Download the License File and Copy It to the DLS Server

→ Go to license server > Action > Download



Download

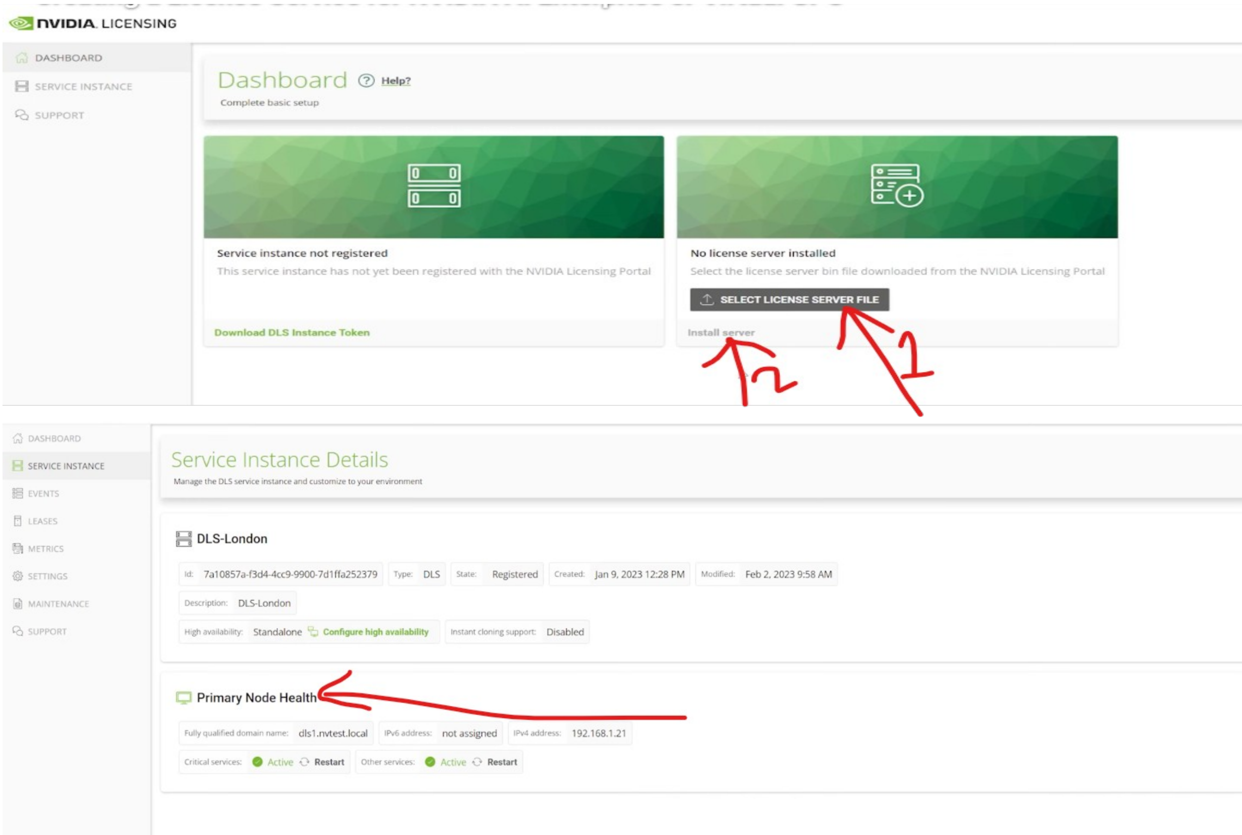


You will receive a fully configured license server file named “LICENSE-xxxx.xx-xxx.bin”, which can be used on the DLS server.

Note: Download .ovf file form nvidia portal of DLS server to install DLS server in you private environment through Vcenter.

10. Install the License Server File on the DLS Server

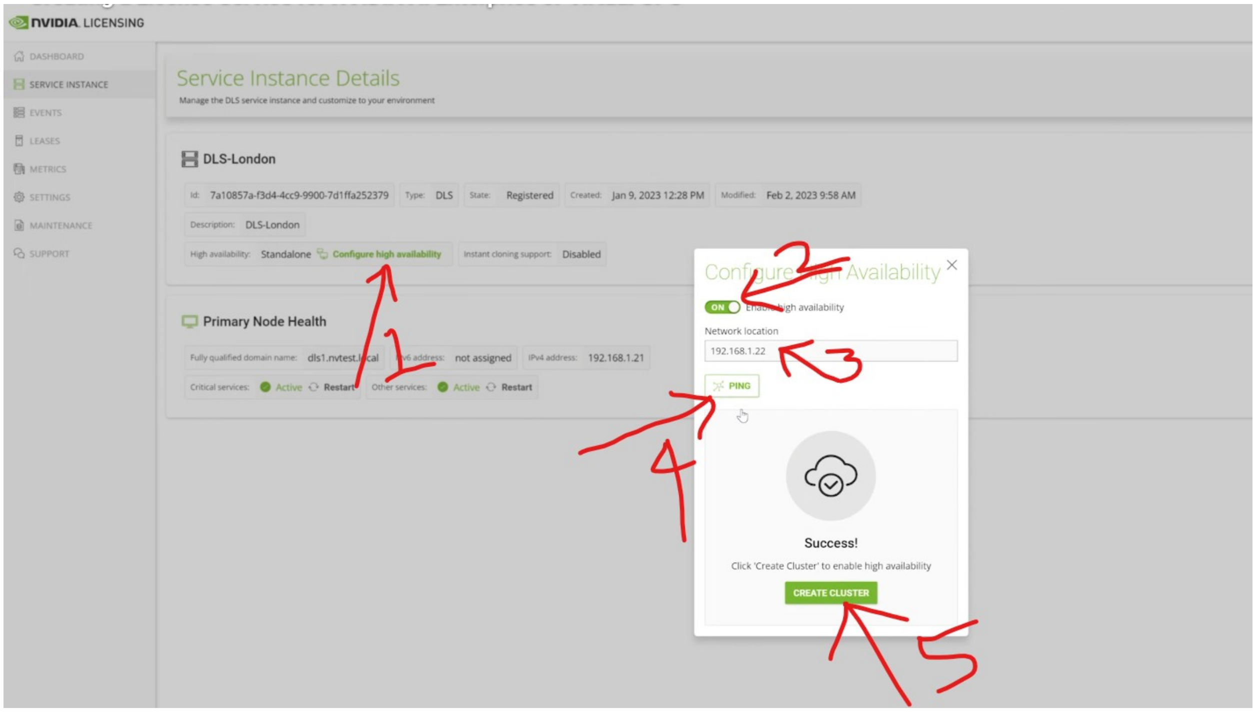
Go to the DLS server dashboard, select the license server file, and click Install.



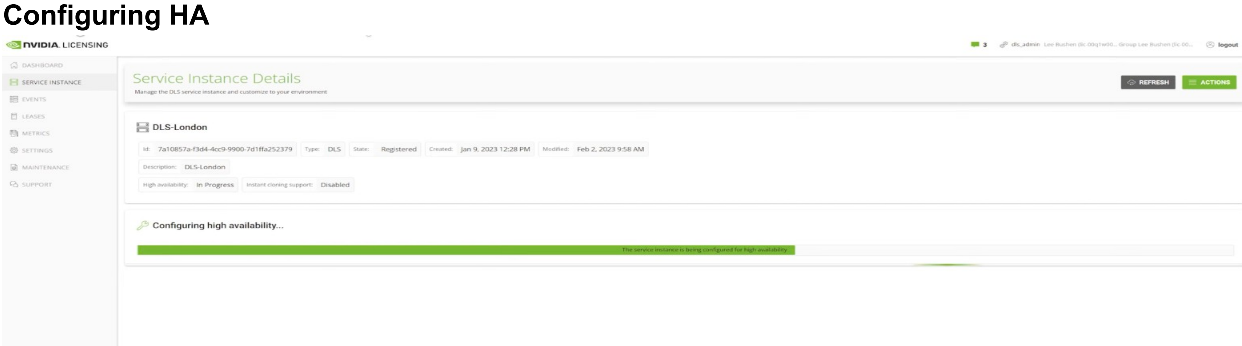
Our license server has been successfully bind to our DLS server

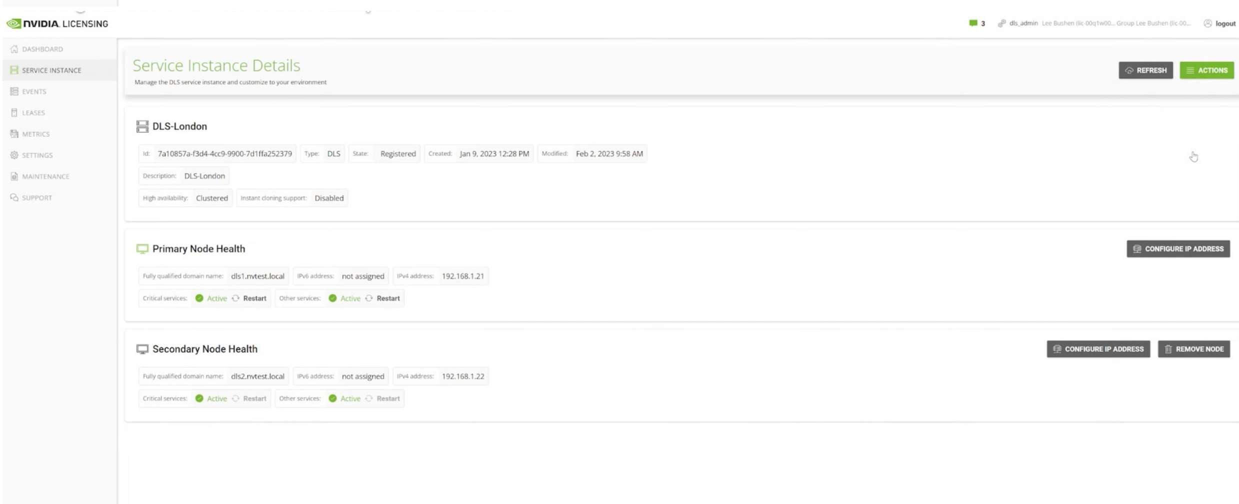
11. Configure DLS High Availability

→ Go to DLS server > Enable HA > Enter Your secondary DLS server IP > Ping > Create server



Configuring HA

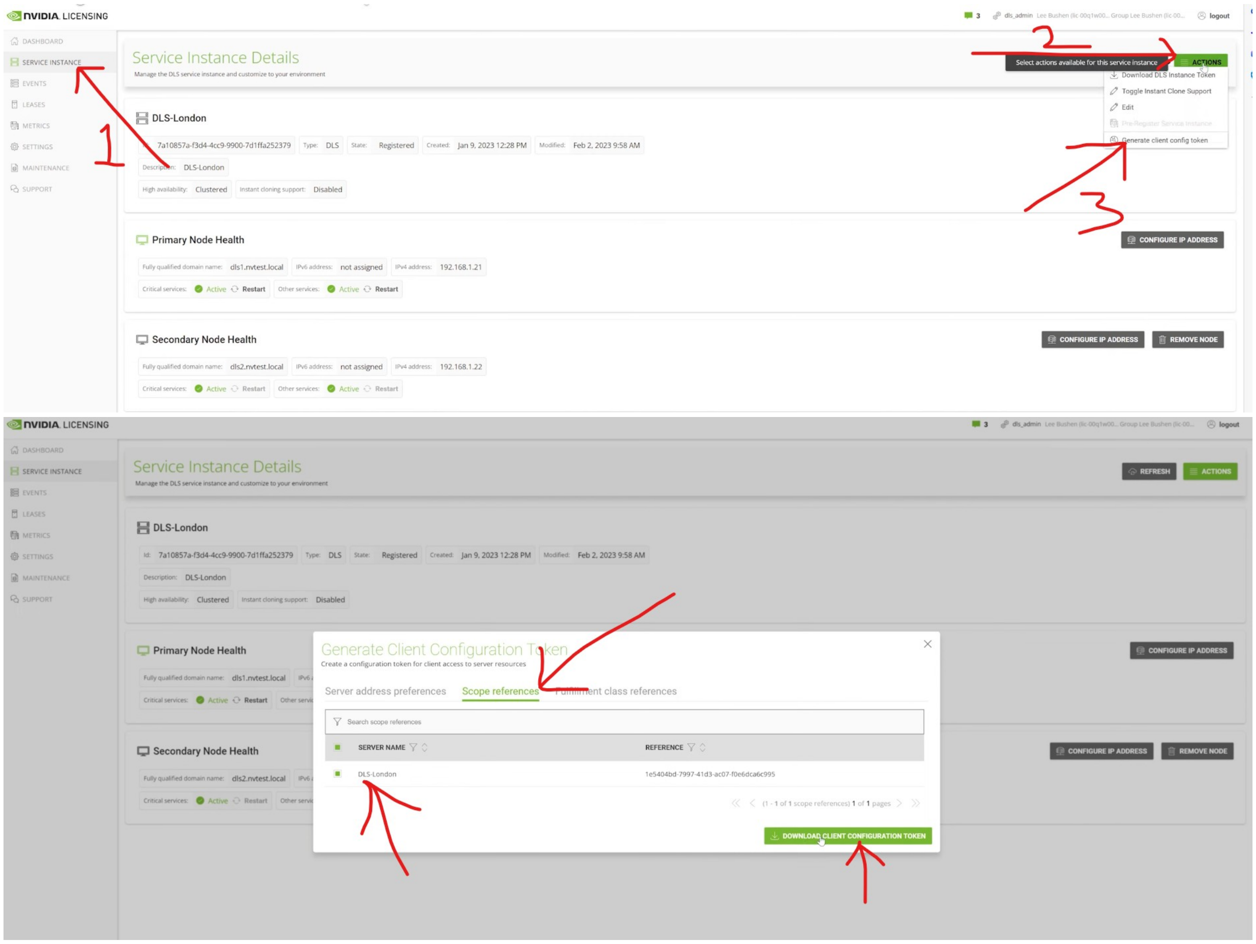




HA configured successfully

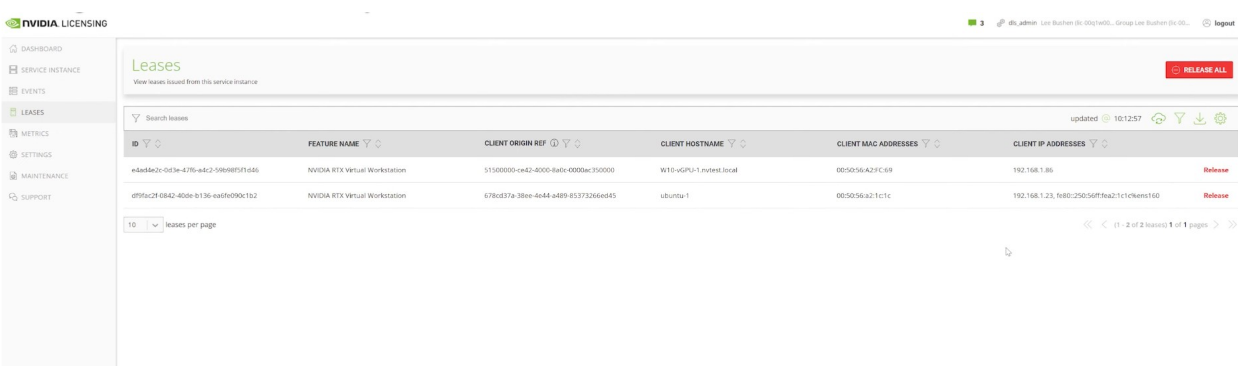
12. Download the Client Configuration Token to assign a license to the virtual machine.

→ Service Instance > Action > Generate Client Config Token > Scope reference > Select server > Download Client Config Token



You will get client_config_token-xx-xx-xx.tok.

13. Verify the VMs on the DLS Server



The DLS license server can manage up to 38k clients.

14. Final Validation Checklist

- ESXi host is in maintenance mode during host-driver installation and reboot.
- NVIDIA host VIB/package is installed and visible in the ESXi software inventory.
- NVIDIA kernel module is loaded.
- nvidia-smi detects the physical GPU on the ESXi host.
- Host Graphics is configured as Shared in vCenter.
- vGPU-related advanced setting is enabled in vCenter.
- A supported VM OS is selected and the required vGPU profile is assigned.
- Secure Boot is disabled for the VM as specified by the supplied procedure.
- Nouveau is disabled and initramfs has been regenerated.
- NVIDIA guest driver is installed and nvidia-smi works inside the VM.
- CUDA Toolkit is installed if required.
- NVIDIA licensing configuration and Client Configuration Token are applied.
- nvidia-gridd service is restarted after token configuration.
- DLS/CLS configuration and HA are completed if used.